Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/22/2005
PI Name:	Whitehead, Joe B. Ph.D.		
Project Title:	Phase Separation and Self-Assembly of Liquid Crystals and Polymer Dispersions: A Ground-Based Feasibility Study for Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Polymers		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	joe.whitehead@usm.edu	**Fax:**	FY 601-266-5149
PI Organization Type:	UNIVERSITY	**Phone:**	601-266-4934
Organization Name:	University of Southern Mississippi		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	Box 5046		
PI Web Page:			
City:	Hattiesburg	**State:**	MS
Zip Code:	39406	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Kaukler , William	**Contact Phone:**	256-544-0693
Contact Email:	William.Kaukler@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):	Crawford, G. P. (Brown University)		
Grant/Contract No.:	NAG8-1684		
Performance Goal No.:			
Performance Goal Text:			

The project constitutes a systematic ground-based investigation of polymerization induced phase separation (PIPS) of liquid crystal and polymer dispersions to elucidate the relationship between process control parameters, the structure properties of the component materials, and the resultant phase- separated morphologies of polymer dispersed liquid crystals (PDLCs) and network stabilized liquid crystals (NSLC). The results will be used to enhance the electro-optic properties of these materials, and as well, to design future reduced-gravity investigations to probe the influence of gravitational acceleration on the phase separation processes. The terrestrial investigation will construct phase diagrams of liquid crystal/monomer and liquid crystal/pre-polymer mixtures based on the type of monomer, pre-polymer, liquid crystal, and the relative amount of each component using thermal analysis, and optical microscopy. In addition, this ground-based investigation includes characterizing micro-structure of PDLC and NSLC materials using thermal analysis, optical microscopy, scanning electron microscopy, voltage dependent laser light transmission, and real-time light

Task Description:	scattering. The design of reduced gravity flight investigation(s) to probe the flow and phase separation of liquid crystal and viscous polymeric materials will be based on the results of ground-based investigation. The objective of this project is to determine the relationships between material properties, processing conditions, and micro-structural evolution of the phase-separated morphology of polymer dispersed liquid crystal (PDLC) and network stabilized liquid crystals (NSLC) materials. Knowledge of the relationships is required to optimize the electro-optic properties of existing material combinations. In addition, the knowledge will lead to new material combinations and processing conditions. LCDPs are fabricated using polymerization induced phase separation (PIPS), Solvent induced phase separation (SIPS), and thermally induced phase separation (TIPS) of liquid crystal/monomer or liquid crystal/pre-polymer mixtures. The microstructure of LCPDs range from that of liquid crystal droplets dispersed in a solid polymer matrix of polymer dispersed liquid crystals (PDLCs) to that of a polymer network dispersed in the liquid crystal of network stabilized liquid crystals (NSLC). LCPDs provide a unique opportunity to investigate the thermodynamics and kinetics of phase transformation from an isotropic liquid to a phase separated anisotropic liquid (liquid crystal) and amorphous polymer. An in-depth understanding of the relationship between the materials, processing conditions, and micro-structure evolution is critical to the design and advancement of PDLC and NSLC materials toward the rigorous requirements for the numerous public and private sector applications that include anti-reflective coatings, omni-directional reflectors, electrically switchable Bragg gratings, spatial light modulators, beam steering, optical interconnects, optical data storage, dynamically variable lenses, and high intensity laser radiation attenuators.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The results of the investigation include the identification of the dominant parameters that control the morphology of PDLCs, and the fabrication of multi-dimensional gratings (lattices) utilizing polymerization induced phase separation (PIPS). The cure temperature and liquid crystal percentage are the two controllable parameters that heavily influence phase separation. The information will lead to better optimization of PDLC and H-PDLC materials and will aid in the design of experiments to probe the phase separation properties in a reduced gravitational environment.
Bibliography Type:	Description: (Last Updated: 08/08/2019)
Articles in Peer-reviewed Journals	Xianyu H, Qi J, Cohn RF, Crawford GP. "Total-internal-reflection mode in holographic polymer dispersed liquid crystals." Opt Lett. 2003 May 15;28(10):792-4. PubMed PMID: 12779148 , May-2003